Ranking Flaky Test Failures with Build Histories and Change Distance

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Flaky tests pass and fail on the same version of the code, and a developer who sees such a failure cannot tell whether the change under review broke the program or whether the test fails at random. Rerunning every failed test costs build time that large projects do not have. This paper ranks the failures of a build by the chance that each failure is flaky. The ranking draws on two signals that every project already records: the history of each test in earlier builds and the distance between the files of the change and the files that the test covers. On 4,210 failures of ten open-source projects, the ranking places a flaky failure first in 71% of the builds that hold one, and a developer who reruns only the three highest ranked failures finds 93% of the flaky ones. The ranking needs no instrumentation and adds less than one second to a build.

CCS Concepts: • **Software and its engineering → Software testing and debugging**.

Additional Key Words and Phrases: flaky tests, regression testing, continuous integration

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1 Introduction

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2 Conclusion

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